

RISANSI INDUSTRIES LTD

# Pump Testing Portal

## Technical Documentation

[rndtesting.risansi.com](http://rndtesting.risansi.com)

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Audience: Developers and maintainers

Status: Internal document

# 1. Overview

testing-portal is a Next.js 15 (App Router) application written in TypeScript. The browser app and the backend live in one repository: pages render in React, and the API is a set of Next.js route handlers under `src/app/api`. Data is stored in an Azure PostgreSQL database (database name `Pump_Selector_and_testing`) accessed through Drizzle ORM and `node-postgres`.

| Item                 | Value                                                                                                                                           |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Framework            | Next.js 15.5 (App Router), React 19, TypeScript                                                                                                 |
| Database access      | Drizzle ORM 0.45 with pg 8 ( <code>node-postgres</code> ) against Azure PostgreSQL                                                              |
| Forms and validation | <code>react-hook-form</code> 7 with <code>zod</code> 4 ( <code>@hookform/resolvers</code> )                                                     |
| Authentication       | Hand-written JWT (HS256) with <code>jsonwebtoken</code> ; passwords hashed with <code>bcryptjs</code>                                           |
| HTTP client          | <code>axios</code> , wrapped in <code>src/services/apiClient.ts</code>                                                                          |
| Styling              | Tailwind CSS v4 utilities mapped to CSS design tokens, plus per-screen CSS files                                                                |
| PDF                  | Server: <code>pdfkit</code> 0.15.2 (pinned) for the audit report. Browser: <code>window.print()</code> with a print stylesheet for test reports |
| Hosting              | Vercel, deployed on every push to main of <a href="https://github.com/farhan2312/risansi-testing">github.com/farhan2312/risansi-testing</a>     |
| Live URL             | <a href="https://rndtesting.risansi.com">https://rndtesting.risansi.com</a>                                                                     |
| Sister app           | <code>sales-portal-next</code> (separate repo and deployment) sharing the users table and JWT secret                                            |

## 2. Architecture

### 2.1 Request flow

- A page file in `src/app/(dashboard)/...` is a thin wrapper that re-exports a screen component from `src/screens/...`
- The screen calls a function in `src/services/*` (`axios`) which calls an API route under `/api`.
- The route handler authenticates the caller from the `auth_token` cookie, checks the role, queries the database through Drizzle, and returns `snake_case` JSON built by the serializers in `src/lib/api.ts`.
- Shared business logic (calculations, permissions, categories) lives in `src/lib` so the UI and the API apply the same rule.

### 2.2 Repository layout

| Path                              | Contents                                                                                                              |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| <code>src/app/(dashboard)/</code> | Thin <code>page.tsx</code> wrappers for every signed-in screen                                                        |
| <code>src/app/api/</code>         | Route handlers: the backend                                                                                           |
| <code>src/screens/</code>         | The real page components (business logic and JSX), one folder per screen                                              |
| <code>src/components/</code>      | ui (modals, headers, pagination), charts (cards, bars, donut, heat map), test-report forms, report-detail, auth guard |

|                                 |                                                                                                 |
|---------------------------------|-------------------------------------------------------------------------------------------------|
| src/layouts/DashboardLayout.tsx | Sidebar, top bar, profile menu, forced-password dialog, notification bell                       |
| src/lib/                        | Auth, DB, calculations, permissions, serializers, date and audit helpers                        |
| private-docs/                   | The documentation PDFs, kept out of /public and served only by /api/admin-docs/[name] to admins |
| src/lib/db/                     | schema.ts (Drizzle tables) and a lazy connection pool                                           |
| src/services/                   | Client-side fetch wrappers                                                                      |
| src/contexts/                   | AuthContext (user held in memory) and ThemeContext                                              |
| src/types/testing.ts            | Shared TypeScript types and constants (categories, statuses, persons, units)                    |
| scripts/                        | One-off .mjs migration and data scripts                                                         |

## 3. Environment, build and run

Configuration is read from .env.local (never committed). Required keys:

| Variable                                        | Purpose                                                                                        |
|-------------------------------------------------|------------------------------------------------------------------------------------------------|
| DB_HOST, DB_PORT, DB_NAME, DB_USER, DB_PASSWORD | PostgreSQL connection                                                                          |
| DB_SSLMODE                                      | require for Azure, disable for local                                                           |
| JWT_SECRET                                      | Signing key. Must be identical to sales-portal-next's JWT_SECRET so a login works in both apps |

The database pool is created lazily through a Proxy, so next build does not fail when credentials are absent at build time.

| Command                                          | What it does                                    |
|--------------------------------------------------|-------------------------------------------------|
| npm run dev                                      | Dev server on http://localhost:5174             |
| npm run build                                    | Production build (includes lint and type-check) |
| npm run start                                    | Serve the build on port 5174                    |
| npx tsc --noEmit                                 | Type-check only. Run after every change         |
| npx dotenv -e .env.local -- node scripts/xxx.mjs | Run a one-off database script                   |
| npm run db:generate / db:migrate / db:push       | drizzle-kit commands                            |

## 4. Authentication and security

- Login (POST /api/auth/login) checks the bcrypt hash and issues a JWT with sub (user id), email, role, iat and exp. Expiry is 12 hours.
- The token is stored in a cookie named auth\_token that is httpOnly, Secure and SameSite=Lax. Browser code never reads it. GET /api/auth/me verifies it and AuthContext keeps the result in memory only.

- decodeToken(req) verifies any valid token. requireAdmin(req) additionally requires role admin. Both reload the live user and reject accounts whose status is not active, so deactivation takes effect immediately.
- Roles: admin, central-admin, source, testing. Permission helpers in src/lib (requisitionPermissions, calendarPermissions) are shared by the API and the UI so the UI never offers what the server refuses.
- Source users only ever see requisitions they created; the rule is enforced in the API queries, not just in the UI.
- Report rules in the API: POST /api/reports refuses source users; PATCH and DELETE on /api/reports/[id] allow the testing role only; PATCH also refuses reports older than 10 days.
- Requisition rules in the API: source users may only PATCH their own requisitions and only the intake fields in SOURCE\_EDITABLE\_FIELDS; DELETE is admin only; closed\_at is set when a non-source user sets status Closed.
- Every account is created with must\_change\_password = true; the layout shows a non-dismissible dialog until it is cleared by a successful password change.
- Destructive actions use the in-app ConfirmModal, never window.confirm, which browsers can silently suppress.
- Deletions, edits and creations are written to audit\_logs under the actor's identity.

## 5. Data model

| Table                   | Purpose and key columns                                                                                                                                                                                                                                                                                  |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| users                   | Accounts: name, email, bcrypt hash, role (admin / central-admin / source / testing), status (pending / active / inactive / rejected), must_change_password, reviewed_by / reviewed_at. Shared with sales-portal-next.                                                                                    |
| test_requisitions       | Testing requests: requisition_no (REQ-nnnnnn), model, category, EC/quotation no., offer date, responsible person, source team, date of requisition, target date, rated QTH / power / head / RPM / capacity (+units), status, retest flags, remarks, created_by, submitted_by (name snapshot), closed_at. |
| pump_test_reports       | Report header: report_no (TR-nnnnnn), model, report_format (observation   viscosity-chart), test type, rated capacity/head/power/RPM, gearbox/motor info, specific gravity, viscosity, K, NPSHa, witness/inspector/recorder, vibration summary, remarks, tested_by, test_date, requisition_id.           |
| pump_test_report_points | One row per test point: head, RPM, capacity, power, volumetric (VE) and mechanical (ME) efficiency and the other computed columns.                                                                                                                                                                       |
| pump_models             | Pump model names used as dropdown suggestions.                                                                                                                                                                                                                                                           |
| requisition_attachments | Files attached to a requisition (metadata plus stored file data; downloaded through a dedicated endpoint).                                                                                                                                                                                               |
| bug_reports             | Bug / feature reports from the "Report a Bug" widget: type, severity, title, description, page, status (Open / In Progress / Resolved), optional screenshot, reporter, is_read.                                                                                                                          |
| audit_logs              | Append-only event trail: user, event_type (login, login_failed, logout, create, update, delete), entity type/id/label, details, IP address, user agent (sign-ins), timestamp.                                                                                                                            |
| user_sessions           | One row per sign-in: login_at, logout_at, last_seen_at, page_view_count.                                                                                                                                                                                                                                 |
| page_views              | One row per page navigation inside a session.                                                                                                                                                                                                                                                            |

|                 |                                                                                                                                                                                |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| action_registry | One row per Assign Retest: unmet fields, rated vs measured snapshot, action points, who assigned, who originally raised the test.                                              |
| calendar_events | Testing Calendar events: title, model, EC/quotation no., responsible person (Sachin   Vikash), status (Planned   Completed), date, optional start/end time, notes, created_by. |

## 5.1 Things to know before touching the schema

- The users table is shared with sales-portal-next. Treat any change to it with care and coordinate; must\_change\_password was added here with an additive script.
- Drizzle's .default() and .\$defaultFn() are application-level for most columns (id, created\_at, updated\_at). Raw SQL inserts must supply id (gen\_random\_uuid()) and timestamps (now()) explicitly.
- Real foreign keys exist that schema.ts does not model: test\_requisitions.created\_by -> users.id and users.reviewed\_by -> users.id. DELETE /api/users/[userId] clears both inside a transaction before deleting; copy that pattern for any other deletion path.
- test\_requisitions.submitted\_by is a name snapshot, not a foreign key, so history stays readable after an account is removed.
- calendar\_events.created\_by is a foreign key to users with ON DELETE SET NULL.
- Migration convention: additive one-off scripts in scripts/ (ALTER TABLE ... ADD COLUMN IF NOT EXISTS) kept as a permanent record; data-fix and debug scripts are deleted after use.

## 6. API reference

All routes are under /api and return JSON with snake\_case field names. "Signed in" means a valid auth\_token cookie. Where a route also applies role rules inside the handler (for example admin-only writes on a route that anyone can read), the purpose column says so. This table is generated from the route files.

| Route                               | Methods | Access    | Purpose                                                             |
|-------------------------------------|---------|-----------|---------------------------------------------------------------------|
| /api/access-requests                | POST    | Public    | Public sign-up request; creates a pending user                      |
| /api/action-registry                | GET     | Signed in | List Assign-Retest entries (Admin / Central Admin)                  |
| /api/admin-docs/[name]              | GET     | Admin     | Admin-only: sends one documentation PDF (inline, or ?download=1)    |
| /api/audit-log/activity             | GET     | Admin     | Paged event log with filters                                        |
| /api/audit-log/export               | GET     | Admin     | Audit report as PDF (pdfkit)                                        |
| /api/audit-log/overview             | GET     | Admin     | Everything the Overview tab draws (adoption, trend, heatmap, grids) |
| /api/audit-log/sessions             | GET     | Admin     | Sign-in sessions list                                               |
| /api/audit-log/summary              | GET     | Admin     | Trailing-24h header figures                                         |
| /api/audit-log/usage                | GET     | Admin     | Per-user usage rollup (sessions, active time, pages)                |
| /api/audit-log/usage/[userId]/pages | GET     | Admin     | Pages one user visited                                              |
| /api/audit/pageview                 | POST    | Signed in | Records a page view for the current session                         |

|                                                   |                    |           |                                                                     |
|---------------------------------------------------|--------------------|-----------|---------------------------------------------------------------------|
| /api/auth/change-password                         | POST               | Signed in | Change own password; clears must_change_password                    |
| /api/auth/login                                   | POST               | Public    | Email + password -> JWT in httpOnly cookie                          |
| /api/auth/logout                                  | POST               | Signed in | Closes the session, clears the cookie                               |
| /api/auth/me                                      | GET                | Signed in | Verifies the cookie and returns the live user                       |
| /api/bug-reports                                  | GET, POST          | Admin     | List / file bug and feature reports                                 |
| /api/bug-reports/[id]                             | GET, PATCH, DELETE | Admin     | Read, change status, delete one report                              |
| /api/bug-reports/[id]/screenshot                  | GET                | Admin     | Download the attached screenshot                                    |
| /api/bug-reports/unread-count                     | GET                | Admin     | Unread count for the sidebar badge                                  |
| /api/calendar-events                              | GET, POST          | Signed in | List events (anyone); create (Admin)                                |
| /api/calendar-events/[id]                         | PATCH, DELETE      | Signed in | Update / delete an event (Admin)                                    |
| /api/calendar-events/ec-options                   | GET                | Signed in | EC / quotation numbers for the event form search                    |
| /api/health                                       | GET                | Public    | Liveness check                                                      |
| /api/overview                                     | GET                | Signed in | Dashboard figures with date, scope, category, model, status filters |
| /api/performance                                  | GET                | Signed in | VE & ME per model against acceptance criteria; improvement list     |
| /api/pump-models                                  | GET                | Signed in | Known pump models (dropdown suggestions)                            |
| /api/pumps                                        | GET                | Signed in | Report Compilation: pump groups, 50 per page                        |
| /api/pumps/[model]                                | GET                | Signed in | Everything for one pump model                                       |
| /api/reports                                      | GET, POST          | Signed in | List / create test reports                                          |
| /api/reports/[id]                                 | GET, PATCH, DELETE | Signed in | Read / edit / delete one report                                     |
| /api/reports/[id]/assign-retest                   | POST               | Signed in | Raise a retest requisition + action-registry row                    |
| /api/reports/grouped                              | GET                | Signed in | Report Archive grouped by pump                                      |
| /api/reports/list                                 | GET                | Signed in | Flat paged report list (drill-downs)                                |
| /api/requisitions                                 | GET, POST          | Signed in | List (filtered, 25/page) / create requisitions                      |
| /api/requisitions/[id]                            | GET, PATCH, DELETE | Admin     | Read / edit / delete one requisition                                |
| /api/requisitions/[id]/attachments                | GET, POST          | Signed in | List / upload attachments                                           |
| /api/requisitions/[id]/attachments/[attachmentId] | GET, DELETE        | Signed in | Download / delete one attachment                                    |
| /api/requisitions/dedup-check                     | GET                | Signed in | Check for earlier tests of the same model/EC                        |

|                                      |               |           |                                                           |
|--------------------------------------|---------------|-----------|-----------------------------------------------------------|
| /api/requisitions/filter-options     | GET           | Signed in | Distinct models, submitters, months for filters           |
| /api/requisitions/target-date-alerts | GET           | Signed in | Items due within 5 days for the bell                      |
| /api/users                           | GET, POST     | Admin     | List users (paged) / add user (Admin)                     |
| /api/users/[userId]                  | PATCH, DELETE | Admin     | Edit role / activate / deactivate / delete a user (Admin) |
| /api/users/[userId]/password         | PATCH         | Admin     | Admin sets a user's password                              |

List endpoints are server-paginated through src/lib/pagination.ts (25 rows per page unless stated) and return { entries, total, page, page\_size }.

## 7. Core business logic modules (src/lib)

| Module                                                   | Responsibility                                                                                                                                                                                              |
|----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| testReportCalc.ts                                        | The live formula engine ported from the Excel workbook (sheets "v notch" and "UPDATE SHEET V-NOTCH (2)"). Only K for Given CPS feeds slip and efficiency; specific gravity and viscosity are informational. |
| requirementCheck.ts                                      | Met / missed for Head, Capacity (reach at least) and Power (ceiling).                                                                                                                                       |
| reportCategory.ts                                        | Derives a report's category from its remarks, EC number or linked requisition.                                                                                                                              |
| reportEnrichment.ts                                      | Adds category, point counts, max VE / ME and unmet fields to report rows.                                                                                                                                   |
| veMeAcceptance.ts                                        | Acceptance table per model, impossible-reading thresholds (VE > 150, ME > 100).                                                                                                                             |
| modelKey.ts                                              | Normalises model names (H20, h-20, H 20) to one key for every lookup.                                                                                                                                       |
| unitConversion.ts                                        | Live converted-unit hints for Capacity and Head on the requisition form. Display only; the stored value is never changed.                                                                                   |
| requisitionAssignment.ts                                 | Default Responsible Person by category (EC Based -> Vikash, otherwise Sachin), applied by the form and enforced by POST /api/requisitions.                                                                  |
| adminDocs.ts                                             | The list of documentation PDFs offered on Admin > Documentation.                                                                                                                                            |
| formUtils.ts                                             | Date formatting and the effective target date (explicit or date + 7 days).                                                                                                                                  |
| reportEditWindow.ts                                      | 10-day edit window.                                                                                                                                                                                         |
| reportDraft.ts, reportFieldMaps.ts                       | Cross-form draft carry-over and snake_case / camelCase field maps.                                                                                                                                          |
| activeTime.ts                                            | Active-time rule: gaps of 15 minutes or less between a user's consecutive actions.                                                                                                                          |
| auditRange.ts                                            | IST date windows for every audit query.                                                                                                                                                                     |
| raisedBy.ts, requisitionFilters.ts, requisitionLookup.ts | Raiser grouping and requisition filter / lookup helpers.                                                                                                                                                    |

## 7.1 Active-time algorithm

Implemented as one SQL derived table in `src/lib/activeTime.ts`. For each user, `audit_logs` rows (excluding `login_failed`) are ordered by time; each row adds the gap to the previous row when that gap is at most 15 minutes (a window function: `lag over partition by user_id`). A gap is credited to the later action's IST day and to the range that contains it. The same derived table feeds Active hours, the Top 3 podium, the per-user day grids, Most active user and the Usage by User tab, so all figures agree.

## 7.2 Dashboard filters

`GET /api/overview` accepts `from`, `to`, `mine`, `category`, `model` and `status`. They are folded into two shared SQL conditions (one for requisitions, one for reports) so every figure, trend and list respects them. A report's category is derived rather than stored, so matching report ids are computed once per request. Reports have no status: `status=Closed` keeps reports, any other status excludes them. Model matching uses the normalised key.

# 8. Frontend

- Pages are client components that load data through services and hold state with React hooks. Forms use `react-hook-form` with zod schemas.
- Infinite scroll (Testing Summary): an `IntersectionObserver` on a sentinel row appends the next 25 rows; guards stop duplicate loads.
- Drill-down modal (`DrillDownModal`): given a `/dashboard?...` or `/reports?...` link it fetches the matching list and renders it in a pop-up. The Overview intercepts clicks on those links.
- Charts are hand-built SVG components (`LineChart`, `StackedBarChart`, `DonutChart`, `Heatmap`, `BarList`) with a table twin for accessibility.
- Theming: design tokens as CSS variables with light and dark palettes; Tailwind classes map to the tokens. The `tw-reset` class scopes a clean button reset to a screen. Print styles force the light palette.
- Do not rely on a `useState(localStorage...)` initialiser for state that must reflect the live session; pair it with a mount effect (see `DashboardLayout`).
- Test report PDFs use `window.print()`. The layout's fixed height and hidden overflow must be reset inside the print media query or exports truncate after one screen.

# 9. Audit and telemetry

- `audit_logs` receives `login`, `login_failed`, `logout` and every `create` / `update` / `delete` made through the API (`logAudit`), with IP address and, for sign-ins, the user agent.
- `user_sessions` is opened at sign-in and closed at sign-out; `/api/audit/pageview` (called on every route change) updates `last_seen_at` and inserts a `page_views` row.
- The Audit Log overview aggregates in one request (`GET /api/audit-log/overview`): adoption (accounts, active, never signed in, dormant, sessions, page views), trend, heat map, per-user grids, devices, browsers and IPs. Day and hour buckets are in IST.

- The audit PDF is generated server-side with pdfkit. Version 0.20 crashes on Vercel, which is why pdfkit is pinned to 0.15.2.

## 10. Deployment and operations

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- Branch main deploys to Vercel automatically on push.
- Database changes: write an additive scripts/add-\*.mjs script, run it with dotenv against the database, commit the script, then deploy the code that uses the new column.
- Both portals read the same database. Take care with changes that touch users.
- A Vercel "Security Checkpoint" page returned to rapid automated requests is bot protection, not an application fault; it clears after a pause.
- Local verification: run the dev server, sign in, and type-check with `npx tsc --noEmit`. There is no automated test framework.

## 11. Conventions and known risks

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- Type-check before every commit; commit and push to main (the live site deploys from it).
- Throwaway scripts and test database rows must be deleted straight after use.
- No automated tests exist; regressions are caught only by manual checking. Adding tests around `testReportCalc`, `requirementCheck` and the audit SQL would be the highest-value first step.
- The shared users table and shared JWT secret couple the two portals; a secret rotation or schema change must be coordinated.
- Active time is conservative because the audit log records sign-ins and writes only. Recording page views as actions, or adding a heartbeat, would raise accuracy.
- Role rules are duplicated by design in API and UI helpers; always change both through the shared helper.